

Модуль 1: "Закалка" Docker (Security & Optimization)

Цель модуля: Перейти от "просто работает" Dockerfile к "безопасному и быстрому" Dockerfile.

1. Задача (Теория): Анализ Dockerfile (Linting).

- **Контекст:** hadolint — это "linter" для Dockerfile.
- **Задание:** Установить hadolint. Написать "плохой" Dockerfile (е.g., RUN apt update и RUN apt install отдельно) и проанализировать его.
- **Критерии:** hadolint показывает ошибки и рекомендации.

```
root@Cetus:/opt/advanced/module1# hadolint --version
Haskell Dockerfile Linter 2.14.0
root@Cetus:/opt/advanced/module1# nano DockerfileBAD
root@Cetus:/opt/advanced/module1# hadolint DockerfileBAD
DockerfileBAD:3 DL3009 info: Delete the apt lists (/var/lib/apt/lists) after installing something
DockerfileBAD:3 DL3027 warning: Do not use apt as it is meant to be an end-user tool, use apt-get or apt-cache instead
DockerfileBAD:4 DL3059 info: Multiple consecutive `RUN` instructions. Consider consolidation.
DockerfileBAD:4 DL3027 warning: Do not use apt as it is meant to be an end-user tool, use apt-get or apt-cache instead
DockerfileBAD:5 DL3027 warning: Do not use apt as it is meant to be an end-user tool, use apt-get or apt-cache instead
DockerfileBAD:5 DL3059 info: Multiple consecutive `RUN` instructions. Consider consolidation.
DockerfileBAD:7 DL3042 warning: Avoid use of cache directory with pip. Use `pip install --no-cache-dir <package>`
DockerfileBAD:7 DL3059 info: Multiple consecutive `RUN` instructions. Consider consolidation.
DockerfileBAD:7 DL3013 warning: Pin versions in pip. Instead of `pip install <package>` use `pip install <package>==<version>` or `pip install --requirement <requirements file>`
root@Cetus:/opt/advanced/module1#
```

2. Задача (Оптимизация): Слияние RUN слоев.

- **Задание:** Исправить Dockerfile из Задачи 1, объединив RUN через && для уменьшения количества слоев.
- **Критерии:** hadolint больше не ругается; docker history показывает меньше слоев.

```
root@Cetus:/opt/advanced/module1# nano DockerfileGOOD
root@Cetus:/opt/advanced/module1# hadolint DockerfileGOOD
DockerfileGOOD:3 DL3027 warning: Do not use apt as it is meant to be an end-user tool, use apt-get or apt-cache instead
DockerfileGOOD:7 DL3013 warning: Pin versions in pip. Instead of `pip install <package>` use `pip install <package>==<version>` or `pip install --requirement <requirements file>`
root@Cetus:/opt/advanced/module1#
```

```
root@Cetus:/opt/advanced/module1# docker history test:good
IMAGE          CREATED          CREATED BY                                      SIZE      COMMENT
f6527bae983a   About a minute ago   RUN /bin/sh -c pip install --no-cache-dir fl...  4.02MB    buildkit.dockerfile.v0
<missing>      About a minute ago   RUN /bin/sh -c apt update && apt install...    43MB      buildkit.dockerfile.v0
<missing>      12 months ago       /bin/sh -c #(nop) CMD ["/bin/bash"]            0B
<missing>      12 months ago       /bin/sh -c #(nop) ADD file:f9ee450324e6ff2c9...  72.8MB
<missing>      12 months ago       /bin/sh -c #(nop) LABEL org.opencontainers....  0B
<missing>      12 months ago       /bin/sh -c #(nop) LABEL org.opencontainers....  0B
<missing>      12 months ago       /bin/sh -c #(nop) ARG LAUNCHPAD_BUILD_ARCH      0B
<missing>      12 months ago       /bin/sh -c #(nop) ARG RELEASE                   0B
root@Cetus:/opt/advanced/module1# docker history test:bad
IMAGE          CREATED          CREATED BY                                      SIZE      COMMENT
84be403bf7a8   18 seconds ago     RUN /bin/sh -c pip install flask # buildkit    4.89MB    buildkit.dockerfile.v0
<missing>      22 seconds ago     RUN /bin/sh -c apt install -y pip # buildkit    297MB     buildkit.dockerfile.v0
<missing>      About a minute ago   RUN /bin/sh -c apt install -y python3 # buil...  39MB      buildkit.dockerfile.v0
<missing>      About a minute ago   RUN /bin/sh -c apt update # buildkit           58.3MB    buildkit.dockerfile.v0
<missing>      12 months ago       /bin/sh -c #(nop) CMD ["/bin/bash"]            0B
<missing>      12 months ago       /bin/sh -c #(nop) ADD file:f9ee450324e6ff2c9...  72.8MB
<missing>      12 months ago       /bin/sh -c #(nop) LABEL org.opencontainers....  0B
<missing>      12 months ago       /bin/sh -c #(nop) LABEL org.opencontainers....  0B
<missing>      12 months ago       /bin/sh -c #(nop) ARG LAUNCHPAD_BUILD_ARCH      0B
<missing>      12 months ago       /bin/sh -c #(nop) ARG RELEASE                   0B
root@Cetus:/opt/advanced/module1#
```

3. Задача (Безопасность): Сканирование уязвимостей (Vulnerability Scanning).

- **Задание:** Установить Trivy (от Aqua Security). Собрать образ на старом python:3.8-slim-buster. Просканировать его: trivy image my-app:1.0.
- **Критерии:** Trivy показывает список CVE (уязвимостей).

```
root@Cetus:/opt/advanced/module1# trivy --version
Version: 0.69.3
```

```
root@Cetus:/opt/advanced/module1# nano DockerfileVULN
root@Cetus:/opt/advanced/module1# docker build -f DockerfileVULN -t my-app:1.0 .
(*) Building 41.3s (10/10) FINISHED
-> [internal] load build definition from DockerfileVULN
-> => transferring dockerfile: 141B
-> [internal] load metadata for docker.io/library/python:3.8-slim-buster
-> [auth] library/python:pull token for registry-1.docker.io
-> [internal] load .dockerignore
-> => transferring context: 2B
-> [1/4] FROM docker.io/library/python:3.8-slim-buster@sha256:8799b8564103a9f36cfb8a8e1c562e11a9a6f2e3bb214e2adc23982b36a04511
-> => resolve docker.io/library/python:3.8-slim-buster@sha256:8799b8564103a9f36cfb8a8e1c562e11a9a6f2e3bb214e2adc23982b36a04511
-> => sha256:8799b8564103a9f36cfb8a8e1c562e11a9a6f2e3bb214e2adc23982b36a04511 988B / 988B
-> => sha256:90834d8a6381dfc3957573dc7a3e6c5c8ed255cf60070329a6da205e6d4257b8 1.37kB / 1.37kB
-> => sha256:addd0962740ab9fd79a788945daa24348c11adcec97d47a647e0a61c86cc9f08 6.87kB / 6.87kB
-> => sha256:8b91b88d557765cd8c6802668755a3f6dc4337b6ce15a17e4857139e5fc964f3 27.14MB / 27.14MB
-> => sha256:82441ee234217961c9c5d4f41df5b295a3c35a671ee52889bf0808e257ec4c 2.78MB / 2.78MB
-> => sha256:8f777578c172d018077d3dc22d6654911fff6066097943fe8c4697ecf8aac35 12.89MB / 12.89MB
-> => sha256:cbfea27189a8b1136859a7973c08243889faf162ebc173a05909dcb0bec03c9 244B / 244B
-> => sha256:276d0cf5deffff3c5d54a8ae8d9a18656a4c03637a8b4f4eef14a147799c901 3.14MB / 3.14MB
-> => extracting sha256:8b91b88d557765cd8c6802668755a3f6dc4337b6ce15a17e4857139e5fc964f3
-> => extracting sha256:82441ee234217961c9c5d4f41df5b295a3c35a671ee52889bf0808e257ec4c
-> => extracting sha256:8f777578c172d018077d3dc22d6654911fff6066097943fe8c4697ecf8aac35
-> => extracting sha256:cbfea27189a8b1136859a7973c08243889faf162ebc173a05909dcb0bec03c9
-> [internal] load build context
-> => transferring context: 512B
-> [2/4] WORKDIR /app
-> [3/4] COPY . .
-> [4/4] RUN pip install flask
-> exporting to image
-> => exporting layers
-> => writing image sha256:0a8bd7a39feca30f24f0519fc9d986976e0092ad77b0e6cc631fee8cd445d
-> => naming to docker.io/library/my-app:1.0
root@Cetus:/opt/advanced/module1# trivy image my-app:1.0
2026-04-08T20:52:52Z INFO [vuln] Need to update DB
2026-04-08T20:52:52Z INFO [vuln] Downloading vulnerability DB...
2026-04-08T20:52:52Z INFO [vuln] Downloading artifact... repo="mirror.gcr.io/aquasec/trivy-db:2"
-----
89.66 MiB / 89.66 MiB [-----]
2026-04-08T20:53:18Z INFO [vuln] Artifact successfully downloaded repo="mirror.gcr.io/aquasec/trivy-db:2"
2026-04-08T20:53:18Z INFO [vul] Vulnerability scanning is enabled
2026-04-08T20:53:18Z INFO [secret] Secret scanning is enabled
2026-04-08T20:53:18Z INFO [secret] If your scanning is slow, please try '--scanners vuln' to disable secret scanning
2026-04-08T20:53:18Z INFO [secret] Please see https://trivy.dev/docs/v0.69/guide/scanner/secret#recommendation for faster secret detection
2026-04-08T20:53:19Z INFO [python] licenses acquired from one or more METADATA files may be subject to additional terms. Use '--debug' flag to see all affected packages.
2026-04-08T20:53:20Z INFO Detected OS family="debian" version="10.13"
2026-04-08T20:53:20Z INFO [debian] Detecting vulnerabilities... os_version="10" pkg_num=93
2026-04-08T20:53:20Z INFO Number of language-specific files num=1
2026-04-08T20:53:20Z INFO [python-pkg] Detecting vulnerabilities...
2026-04-08T20:53:20Z WARN Using severities from other vendors for some vulnerabilities. Read https://trivy.dev/docs/v0.69/guide/scanner/vulnerability#severity-selection for details.
2026-04-08T20:53:20Z WARN This OS version is no longer supported by the distribution family="debian" version="10.13"
2026-04-08T20:53:20Z WARN The vulnerability detection may be insufficient because security updates are not provided
2026-04-08T20:53:20Z INFO Table result includes only package filenames. Use '--format json' option to get the full path to the package file.

Report Summary
```

Target	Type	Vulnerabilities	Secrets
my-app:1.0 (debian 10.13)	debian	125	-
usr/local/lib/python3.8/site-packages/MarkupSafe-2.1.5.dist-info/METADATA	python-pkg	0	-
usr/local/lib/python3.8/site-packages/blinker-1.8.2.dist-info/METADATA	python-pkg	0	-
usr/local/lib/python3.8/site-packages/click-8.1.8.dist-info/METADATA	python-pkg	0	-
usr/local/lib/python3.8/site-packages/flask-3.0.3.dist-info/METADATA	python-pkg	1	-

my-app:1.0 (debian 10.13)

Total: 125 (UNKNOWN: 3, LOW: 14, MEDIUM: 66, HIGH: 40, CRITICAL: 2)

4. Задача (Исправление): Пересобрать образ на python:3.11-slim-bullseye (более новой).

- Критерии: trivy image my-app:2.0 показывает 0 HIGH/CRITICAL уязвимостей.

```
root@Cetus:/opt/advanced/module1# trivy image my-app:1.1
2026-04-08T20:57:22Z INFO [vuln] Vulnerability scanning is enabled
2026-04-08T20:57:22Z INFO [secret] Secret scanning is enabled
2026-04-08T20:57:22Z INFO [secret] If your scanning is slow, please try '--scanners vuln' to disable secret scanning
2026-04-08T20:57:22Z INFO [secret] Please see https://trivy.dev/docs/v0.69/guide/scanner/secret#recommendation for faster secret detection
2026-04-08T20:57:22Z INFO [python] licenses acquired from one or more METADATA files may be subject to additional terms. Use '--debug' flag to see all affected packages.
2026-04-08T20:57:22Z INFO Detected OS family="debian" version="11.11"
2026-04-08T20:57:22Z INFO [debian] Detecting vulnerabilities... os_version="11" pkg_num=104
2026-04-08T20:57:22Z INFO Number of language-specific files num=1
2026-04-08T20:57:22Z INFO [python-pkg] Detecting vulnerabilities...
2026-04-08T20:57:22Z WARN Using severities from other vendors for some vulnerabilities. Read https://trivy.dev/docs/v0.69/guide/scanner/vulnerability#severity-selection for details.
2026-04-08T20:57:22Z INFO Table result includes only package filenames. Use '--format json' option to get the full path to the package file.

Report Summary
```

Target	Type	Vulnerabilities	Secrets
my-app:1.1 (debian 11.11)	debian	205	-
usr/local/lib/python3.11/site-packages/blinker-1.9.0.dist-info/METADATA	python-pkg	0	-
usr/local/lib/python3.11/site-packages/click-8.3.2.dist-info/METADATA	python-pkg	0	-
usr/local/lib/python3.11/site-packages/flask-3.1.3.dist-info/METADATA	python-pkg	0	-
usr/local/lib/python3.11/site-packages/itsdangerous-2.2.0.dist-info/METADATA	python-pkg	0	-
usr/local/lib/python3.11/site-packages/jinja2-3.1.6.dist-info/METADATA	python-pkg	0	-
usr/local/lib/python3.11/site-packages/markupsafe-3.0.3.dist-info/METADATA	python-pkg	0	-
usr/local/lib/python3.11/site-packages/pip-24.0.dist-info/METADATA	python-pkg	2	-
usr/local/lib/python3.11/site-packages/setuputils-65.5.1.dist-info/METADATA	python-pkg	2	-
usr/local/lib/python3.11/site-packages/werkzeug-3.1.8.dist-info/METADATA	python-pkg	0	-
usr/local/lib/python3.11/site-packages/wheel-0.45.1.dist-info/METADATA	python-pkg	1	-

Legend:
- '-': Not scanned
- '0': Clean (no security findings detected)

my-app:1.1 (debian 11.11)


```
my-app:1.1 (debian 11.11)
```

```
Total: 205 (UNKNOWN: 6, LOW: 111, MEDIUM: 63, HIGH: 22, CRITICAL: 3)
```

5. Задача (Теория): BuildKit.

- Контекст: BuildKit — это *новый*, параллельный и более быстрый сборщик Docker.
- Задание: Включить BuildKit (DOCKER_BUILDKIT=1). Пересобрать образ.
- Критерии: Вывод docker build изменился (стал "древовидным").

```
root@Cetus:/opt/advanced/module1# export DOCKER_BUILDKIT=1
root@Cetus:/opt/advanced/module1# docker build -t my-app:bk .
[+] Building 0.2s (1/1) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 2B
ERROR: failed to build: failed to solve: failed to read dockerfile: open Dockerfile: no such file or directory
root@Cetus:/opt/advanced/module1# ls
DockerfileBAD DockerfileFIXED DockerfileGOOD DockerfileVULN
root@Cetus:/opt/advanced/module1# docker build -f DockerfileFIXED -t my-app:bk .
[+] Building 1.5s (10/10) FINISHED
=> [internal] load build definition from DockerfileFIXED
=> => transferring dockerfile: 160B
=> [internal] load metadata for docker.io/library/python:3.11-slim-bullseye
=> [auth] library/python:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/4] FROM docker.io/library/python:3.11-slim-bullseye@sha256:9e25f400253a5fa3191813d6a67eb801ca1e6f012b3bd2588fa6920b59e3eba6
=> [internal] load build context
=> => transferring context: 135B
=> CACHED [2/4] WORKDIR /app
=> CACHED [3/4] COPY . .
=> CACHED [4/4] RUN pip install --no-cache-dir flask
=> exporting to image
=> => exporting layers
=> => writing image sha256:c4c8863cd917b3de41286424de7f135be99614ec91b3246967b00308c80fdc62
=> => naming to docker.io/library/my-app:bk
root@Cetus:/opt/advanced/module1#
```

6. Задача (BuildKit Cache): Использование RUN --mount=type=cache.

- Контекст: Как кешировать apt или pip между сборками?
- Задание: Изменить RUN pip install ... на RUN --mount=type=cache,target=/root/.cache/pip pip install
- Критерии: Повторная сборка (даже после docker builder prune) использует кеш pip, что ускоряет сборку.

```
root@Cetus:/opt/advanced/module1# echo "flask" > requirements.txt
root@Cetus:/opt/advanced/module1# nano DockerfileCACHE
root@Cetus:/opt/advanced/module1# root@Cetus:/opt/advanced/module1# docker build -f DockerfileCACHE -t my-app:cache .
[+] Building 23.0s (10/10) FINISHED
=> [internal] load build definition from DockerfileCACHE
=> => transferring dockerfile: 235B
=> [internal] load metadata for docker.io/library/python:3.11-slim-bullseye
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [stage-0 1/5] FROM docker.io/library/python:3.11-slim-bullseye@sha256:9e25f400253a5fa3191813d6a67eb801ca1e6f012b3bd2588fa6920b59e3eba6
=> [internal] load build context
=> => transferring context: 221B
=> CACHED [stage-0 2/5] WORKDIR /app
=> [stage-0 3/5] COPY requirements.txt .
=> [stage-0 4/5] RUN --mount=type=cache,target=/root/.cache/pip pip install -r requirements.txt
=> [stage-0 5/5] COPY . .
=> exporting to image
=> => exporting layers
=> => writing image sha256:20c14ec19e884ba803b77754b34753a31eb62837a0f68c2bcbb328294b136ca
=> => naming to docker.io/library/my-app:cache
root@Cetus:/opt/advanced/module1# docker builder prune
WARNING! This will remove all dangling build cache. Are you sure you want to continue? [y/N] y
ID                RECLAIMABLE  SIZE    LAST ACCESSED
e5nkeykrkgd5egin696zeg62s  true         631B    2 months ago
od24a7sngdh5b3w1dq8fa8x45*  true        438.1kB  2 months ago
uy9ut80gdaa5naee6f2xjisg*  true        437.8kB  2 months ago
```



```

root@Cetus:/opt/advanced/module1# docker buildx build \
> --platform linux/amd64,linux/arm64 \
> -f DockerfileFIXED \
> -t yelopukki/my-app:latest \
> --push .
[+] Building 32.1s (15/15) FINISHED
=> [internal] load build definition from DockerfileFIXED
=> => transferring dockerfile: 160B
=> [linux/arm64 internal] load metadata for docker.io/library/python:3.11-slim-bullseye
=> [linux/amd64 internal] load metadata for docker.io/library/python:3.11-slim-bullseye
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [linux/amd64 1/4] FROM docker.io/library/python:3.11-slim-bullseye@sha256:9e25f400253a5fa3191813d6a67eb801ca1e6f012b3bd2588fa6920b59e3eba6
=> => resolve docker.io/library/python:3.11-slim-bullseye@sha256:9e25f400253a5fa3191813d6a67eb801ca1e6f012b3bd2588fa6920b59e3eba6
=> [linux/arm64 1/4] FROM docker.io/library/python:3.11-slim-bullseye@sha256:9e25f400253a5fa3191813d6a67eb801ca1e6f012b3bd2588fa6920b59e3eba6
=> => resolve docker.io/library/python:3.11-slim-bullseye@sha256:9e25f400253a5fa3191813d6a67eb801ca1e6f012b3bd2588fa6920b59e3eba6
=> [internal] load build context
=> => transferring context: 205B
=> CACHED [linux/amd64 2/4] WORKDIR /app
=> CACHED [linux/amd64 3/4] COPY . .
=> CACHED [linux/amd64 4/4] RUN pip install --no-cache-dir flask
=> CACHED [linux/arm64 2/4] WORKDIR /app
=> CACHED [linux/arm64 3/4] COPY . .
=> CACHED [linux/arm64 4/4] RUN pip install --no-cache-dir flask
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:961cc85a35d9cdc94d1c86b32d6267cd7180a8a793072c83f6757a5c73c8cd6e
=> => exporting config sha256:27dc75badfee7a8eaddc8314973ef7dd394cc734bcb79bf9d6baa03f16e7f89d
=> => exporting attestation manifest sha256:8d348bdf8ce323f418eb9c72f36b5c495558ca0b688f744feaa47024f549e70
=> => exporting manifest sha256:8b9222f1c832f34dc511589fb865cc2f9e5add114193c927290d4bd83d19803c
=> => exporting config sha256:70a950e0f0d2494cb5907eb2ab050ca907341592167d1fc4fe2a61269076a985
=> => exporting attestation manifest sha256:75cb87043967e7fd4cec42e0c1edcb3bdf76bfcea2777bd317b9a030a25a5fe
=> => exporting manifest list sha256:0367b9fbbfa99ea06e40bde28d4212a8abe74294e3761e04584cad4ef9be313c
=> => pushing layers
=> => pushing manifest for docker.io/yelopukki/my-app:latest@sha256:0367b9fbbfa99ea06e40bde28d4212a8abe74294e3761e04584cad4ef9be313c
=> [auth] yelopukki/my-app:pull,push token for registry-1.docker.io
root@Cetus:/opt/advanced/module1#

```

TAG			
latest			
Last pushed less than a minute by yelopukki			
		docker pull yelopukki/my-app:latest	
Digest	OS/ARCH	Last pull	Compressed size
961cc85a35d9	linux/amd64	less than 1 day	49.99 MB
8b9222f1c832	linux/arm64	less than 1 day	48.58 MB

8. Задача (Build Secrets): Безопасная передача токенов во время сборки.

- **Контекст:** Как использовать приватный `pip` репозиторий?
- **Задание:** Изучить `RUN --mount=type=secret,id=mysecret ....`
- **Критерии:** Секрет (e.g., `pip.conf`) используется при сборке, но не сохраняется в слоях образа.


```

root@Cetus:/opt/advanced/module1# DOCKER_BUILDKIT=1 docker build -f DockerfileSECRET --secret id=mysecret,src=secret.txt -t secret-test .
[+] Building 19.9s (7/7) FINISHED
=> [internal] load build definition from DockerfileSECRET
=> => transferring dockerfile: 134B
=> [internal] load metadata for docker.io/library/python:3.11-slim
=> [auth] library/python:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [stage-0 1/2] FROM docker.io/library/python:3.11-slim@sha256:b1b81d67b8df73bf6067191790c233ef9e598b863fc1ccf447956138ca466d99
=> => resolve docker.io/library/python:3.11-slim@sha256:b1b81d67b8df73bf6067191790c233ef9e598b863fc1ccf447956138ca466d99
=> => sha256:d1a45c0fb43d72e0c013ee97d460aa8f49f2c1bbd588b17a558cc6f7078b4276 5.48kB / 5.48kB
=> => sha256:5435b2dcdf5cb7faa0d5b1d4d54be2c72a776fab9a605336f5067d6e9ecb5976 29.78MB / 29.78MB
=> => sha256:37ffa6577b04e98598e2c10432eb8c66cfc1af6c7698b21c6163f8334606aa14 1.29MB / 1.29MB
=> => sha256:d98f36ba21593a4014243f41ec9c5370abd34f4f65a5d074e41bce23f5f6a165 14.37MB / 14.37MB
=> => sha256:b1b81d67b8df73bf6067191790c233ef9e598b863fc1ccf447956138ca466d99 10.37kB / 10.37kB
=> => sha256:ad65166ad2583036804a3f772ec8f1cad733cf77d631c6840ab1ea9231b6b350 1.75kB / 1.75kB
=> => sha256:0016ee351b60c03c7a77a3f50c717891eedd927f4e6a1d59f6cce05268065606 251B / 251B
=> => extracting sha256:5435b2dcdf5cb7faa0d5b1d4d54be2c72a776fab9a605336f5067d6e9ecb5976
=> => extracting sha256:37ffa6577b04e98598e2c10432eb8c66cfc1af6c7698b21c6163f8334606aa14
=> => extracting sha256:d98f36ba21593a4014243f41ec9c5370abd34f4f65a5d074e41bce23f5f6a165
=> => extracting sha256:0016ee351b60c03c7a77a3f50c717891eedd927f4e6a1d59f6cce05268065606
=> [stage-0 2/2] RUN --mount=type=secret,id=mysecret cat /run/secrets/mysecret
=> exporting to image
=> => exporting layers
=> => writing image sha256:23dd4790784e7c47ee90a09dd56071b49ac797ed5086e13743b499f4c7e5a1da
=> => naming to docker.io/library/secret-test
root@Cetus:/opt/advanced/module1# docker history secret-test
IMAGE          CREATED              CREATED BY                                      SIZE      COMMENT
23dd4790784e   35 seconds ago      RUN /bin/sh -c cat /run/secrets/mysecret # b... 0B         buildkit.dockerfile.v0
<missing>      43 hours ago        CMD ["python3"]                                0B         buildkit.dockerfile.v0
<missing>      43 hours ago        RUN /bin/sh -c set -eux; for src in idle3 p... 36B        buildkit.dockerfile.v0
<missing>      43 hours ago        RUN /bin/sh -c set -eux; savedAptMark="$(a... 42MB        buildkit.dockerfile.v0
<missing>      43 hours ago        ENV PYTHON_SHA256=272179dd9a2e41a0fc8e42e33... 0B          buildkit.dockerfile.v0
<missing>      43 hours ago        ENV PYTHON_VERSION=3.11.15                    0B          buildkit.dockerfile.v0
<missing>      43 hours ago        ENV GPG_KEY=A035C8C19219BA821ECEA86B64E628F... 0B          buildkit.dockerfile.v0
<missing>      43 hours ago        RUN /bin/sh -c set -eux; apt-get update; a... 3.81MB      buildkit.dockerfile.v0
<missing>      43 hours ago        ENV LANG=C.UTF-8                               0B          buildkit.dockerfile.v0
<missing>      43 hours ago        ENV PATH=/usr/local/bin:/usr/local/sbin:/usr... 0B          buildkit.dockerfile.v0
<missing>      2 days ago          # debian.sh --arch 'amd64' out/ 'trixie' '@1... 78.6MB      debuerreotype 0.17
root@Cetus:/opt/advanced/module1# docker run -it --rm secret-test sh
# ls /run/secrets
ls: cannot access '/run/secrets': No such file or directory
#

```

9. Задача (Slim-образы): Использование distroless (Google).

- **Задание:** Написать multi-stage Dockerfile (е.g., для Go/Java), где final образ — gcr.io/distroless/static (не содержит *ничего*, кроме вашего бинарника).
- **Критерии:** Итоговый образ весит < 20 МБ (для Go). docker run ... bash не работает (т.к. bash в образе нет).

```

root@Cetus:/opt/advanced/module1# docker build -f DockerfileDIST -t distroless-app .
[+] Building 7.7s (12/12) FINISHED
=> [internal] load build definition from DockerfileDIST
=> => transferring dockerfile: 220B
=> [internal] load metadata for gcr.io/distroless/static:latest
=> [internal] load metadata for docker.io/library/golang:1.22
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [builder 1/4] FROM docker.io/library/golang:1.22@sha256:1cf6c45ba39db9fd6db16922041d074a63c93556a85c5ccb62d181034df7f02
=> CACHED [stage-1 1/2] FROM gcr.io/distroless/static:latest@sha256:47b2d72ff90843eb8a768b5c2f89b40741843b639d865b9b937b07cd59b479c6
=> [internal] load build context
=> => transferring context: 450B
=> CACHED [builder 2/4] WORKDIR /app
=> [builder 3/4] COPY .
=> [builder 4/4] RUN go build -o app
=> [stage-1 2/2] COPY --from=builder /app/app /app
=> exporting to image
=> => exporting layers
=> => writing image sha256:7b1e4cd6c058a2e5e2c714f949c104676dfab1e5abb809b47317dd02a8209ab0
=> => naming to docker.io/library/distroless-app
root@Cetus:/opt/advanced/module1# docker run distroless-app
Hello from distroless!
root@Cetus:/opt/advanced/module1# docker run distroless-app
Hello from distroless!
root@Cetus:/opt/advanced/module1# docker run -it distroless-app bash
docker: Error response from daemon: failed to create task for container: failed to create shim task: OCI runtime create failed: runc create failed: unable to start container process: error
Run 'docker run --help' for more information

```

```

root@Cetus:/opt/advanced/module1# docker images

```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
my-app	latest	8d04fc1e353d	24 minutes ago	143MB
<none>	<none>	5a4b8772086e	26 minutes ago	143MB
<none>	<none>	0ae926adc4d1	29 minutes ago	143MB
distroless-app	latest	7b1e4cd6c058	33 minutes ago	4.11MB
secret-test	latest	23dd4790784e	49 minutes ago	124MB
my-app	cache	20c14ec19e88	About an hour ago	143MB

10. Задача (Docker Compose Override): Разделение dev и prod в Compose.

- **Задание:** Создать docker-compose.yml (для prod, использует image: my-app:latest) и docker-compose.override.yml (для dev, использует build: . и volumes: [./app]).
- **Критерии:** docker-compose up (запускает dev), docker-compose -f docker-compose.yml up (запускает prod).

```
root@Cetus:/opt/advanced/module1# docker compose up -d
WARN[0000] /opt/advanced/module1/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
WARN[0000] /opt/advanced/module1/docker-compose.override.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Running 2/2
   █ Network module1_default Created
   █ Container module1-app-1 Started
root@Cetus:/opt/advanced/module1# curl http://localhost:5000
Hello DEV!root@Cetus:/opt/advanced/module1# nano app.py
root@Cetus:/opt/advanced/module1# curl http://localhost:5000
Hello DEVDEVDEVDEVDEV!root@Cetus:/opt/advanced/module1#
```

```
root@Cetus:/opt/advanced/module1# docker-compose -f docker-compose.yml up -d
Starting module1_app_1 ... done
root@Cetus:/opt/advanced/module1# docker ps
CONTAINER ID   IMAGE                COMMAND                  CREATED        STATUS        PORTS                    NAMES
e92d92212ec5   my-app:latest        "python app.py"         48 seconds ago Up 5 seconds   0.0.0.0:5000->5000/tcp   module1_app_1
24df5703917e   moby/buildkit:buildx-stable-1 "/usr/bin/buildkitd-..." 10 minutes ago Up 10 minutes                               buildx_buildkit_multi0
root@Cetus:/opt/advanced/module1# docker-compose down
Stopping module1_app_1 ... done
Removing module1_app_1 ... done
Removing network module1_default
```